

Nathaniel Martes

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EDUCATION

B.S. in Computer Science, Wilkes University, GPA: 3.95/4.00, Honors

Expected Graduation – May 2027

EXPERIENCE

Software Engineer Intern, NVIDIA

May 2026 – Sep 2026

- Implemented features in an open-source distributed storage system, OpenStack Swift, using Linux, Git, and Python, which supports 1+ Exabyte (EB) of storage and autonomous vehicle AI workflows generating 160M+ requests per day.
- Architected and developed a Python middleware replicating AWS S3 checksums, enabling data workflows handling 41 Petabytes (PB) to leverage 90% faster checksum algorithms, ensuring API compatibility through Pytest functional tests.
- Integrated AI agent workflows using tools like OpenAI's Codex and Anthropic's Claude Code, cutting development time for new storage features and tests and accelerating release readiness.
- Utilized Prometheus, Grafana, and OpenSearch to analyze system metrics, which identified 700+ client requests at risk from upcoming updates, enabling data-driven production decisions and reducing deployment risk.
- Accelerated provisioning and teardown of distributed system object storage benchmarking environments using Python, Ansible and Linux, reducing deployment time and improving testing efficiency.
- Reviewed 20+ open-source changes, identifying design improvements, leading to better code quality before merge.
- Partnered with cross-functional teams to communicate software release changes, improving stakeholder visibility.

Software Engineer Intern, VetsEZ

Sep 2025 – May 2026

- Built a REST API microservice with a development team using Java, SQL, Spring Boot, Git, and Postman, which enabled platform modernization efforts by exposing legacy data schemas.
- Reduced service latency by 3 minutes using a metadata cache created with Java, Docker, AWS ECS, and PostgreSQL.
- Engineered unit tests into a GitHub Actions CI/CD pipeline for a SwaggerUI OpenAPI documentation generator using Java, successfully increasing generator code coverage to 91%.
- Automated REST API RPC regression testing with Python and Postman, saving developers 10 hours of manual testing.
- Cooperated with developers using Microsoft Office and Github Issues, improving code quality and coherence.

Software Engineer Intern, Software Engineering Institute at Carnegie Mellon University

May 2025 – Aug 2025

- Applied Python, Git, Docker, and REST APIs to build a vulnerability detector for installed packages across Red Hat Linux (RHEL) systems, reducing detection time from 30 days to 1 day.
- Corrected 4,400+ SQLite entries with Python, achieving 100% accuracy in quarterly software license usage reports.
- Developed a Python REST API tool to detect duplicate software license usage, saving 2 hours of manual data analysis.
- Deployed 80+ test cases into DevOps CI/CD pipelines for Jira, Confluence, Bamboo, and Bitbucket instances using Python, Pytest, and REST APIs, preventing deployment of misconfigured Docker containers.
- Collaborated in an Agile environment using Jira to track project progress and participate in daily stand-up meetings, enhancing team communication and improving overall project understanding.

System Administrator Intern, Wilkes University

Jun 2024 – Current

- Optimized Linux Ubuntu server administration tasks using Ansible, saving 1 hour during software updates.
- Developed a script with Bash to wake managed workstations, allowing administrations to update systems remotely.
- Configured Python toolchains on workstations using Ansible, making it easier for students to debug programs.
- Migrated LDAP to LDAPS using Bash, Ansible, and Linux, strengthening protection of user credentials in transit.

PROJECTS

AWS Affiliated Hackathon 1st Place, Interactive AI Reflection Therapy Web Application

Oct 2025 – Oct 2025

- Created a serverless full stack application using JavaScript, React, HTML, Tailwind CSS, Python, Git, and DynamoDB that creates AI interactive narrative chapters for therapy and user resonance which placed 1st among 200+ participants.
- Utilized AWS Cloud Development Kit (CDK) to create infrastructure-as-code (IoC) which deployed a REST API Gateway and Lambda functions that interacted with a NoSQL database and a Large Language Model (LLM).
- Collaborated with 2 developers to design, develop, and deploy the complete application within a 24-hour period.

J.P. Morgan Chase Affiliated Hackathon 1st Place, AI Image Disaster Analyzer

Mar 2025 – Mar 2025

- Worked in a team of 4 to develop a full stack AI image analysis application for determining room disaster readiness using Python, JavaScript, TypeScript, React, HTML, and CSS which placed 1st among 470+ participants.
- Built a REST API using FastAPI and Google Gemini to perform server side prompting and AI parsing.
- Established Git and Docker development workflows which lead to better team efficiency under limited time.

LEADERSHIP

President, Society of Professional Hispanic Engineers (SHPE) Wilkes University Chapter

May 2026 – Current

- Secured \$3800+ in funding by presenting a convention funding proposal to 15+ student government members.
- Promoted SHPE across campus through recruitment efforts, bringing in 30+ new members.
- Managed the Wilkes SHPE Chapter's LinkedIn presence, highlighting student achievements, community involvement, and chapter updates to an audience of 850+ followers.

Vice President, Mathematics and Computer Science Club at Wilkes University

Jan 2025 – Current

- Represented the student body at the dedication of a \$4.3M donation, delivering remarks on its impact on academic facilities, technology, and future student opportunities.
- Managed Discord communications for 60+ club members, sharing meeting information, chapter announcements, and university events.

Martial Arts Instructor, Dual Dragon Karate

May 2022 – Current

- Led hundreds of classes teaching self-defense techniques and traditional forms to 15+ students.
- Handled different student age groups from 4 years old to 50+, ensuring students are satisfied with the class and that they are learning real world skills.

SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, Go, SQL

Frameworks & Libraries: Spring Boot, FastAPI, React, Pytest

Cloud: AWS, Lambda, Amazon S3, EC2, VPC, RDS, ECS, CloudFormation

Infrastructure: Kubernetes, Docker, Terraform, Kafka, Ansible, CI/CD

Databases & Data: PostgreSQL, JSON, YAML

Developer Tools: Git, Linux, Virtual Machines, HTML, CSS